PalmSource Developer Conference 2004
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Palm OS® Development Tools

Overview and Roadmap

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palm POWERED

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What's new?

- **EVERYTHING!**
  - IDE
  - Compilers
  - Debuggers
  - Resource Editor
  - Resource Compiler
- **Well, almost everything**
  - Still 68K applications shipping
  - Still PNO applications shipping



Why is everything new?

- **Native ARM Development**
- **New set of APIs**
- **Whole new underpinnings**



Integrated Development Environment

- **Based on Eclipse**
- **State-of-the-art Java Development Tools**
- **Up-and-coming C/C++ Development Tools**
- **Open Architecture**
- **Powerful plug-in capabilities**
- **Liberal open source licensing (EPL v.1.0)**
- **Large number of high quality committers**
- **Larger number of available plug-ins**

(See Eclipse Session, Thursday 1:30pm)



Integrated Development Environment

- **Palm OS® Developer Suite**
  - Palm OS 68K Application Development
  - Palm OS PNO/68K Application Development
  - Palm OS 6 Application Development
  - Support for C and C++ Development
  - Application template generation (Wizards)
  - Support for Software Updates



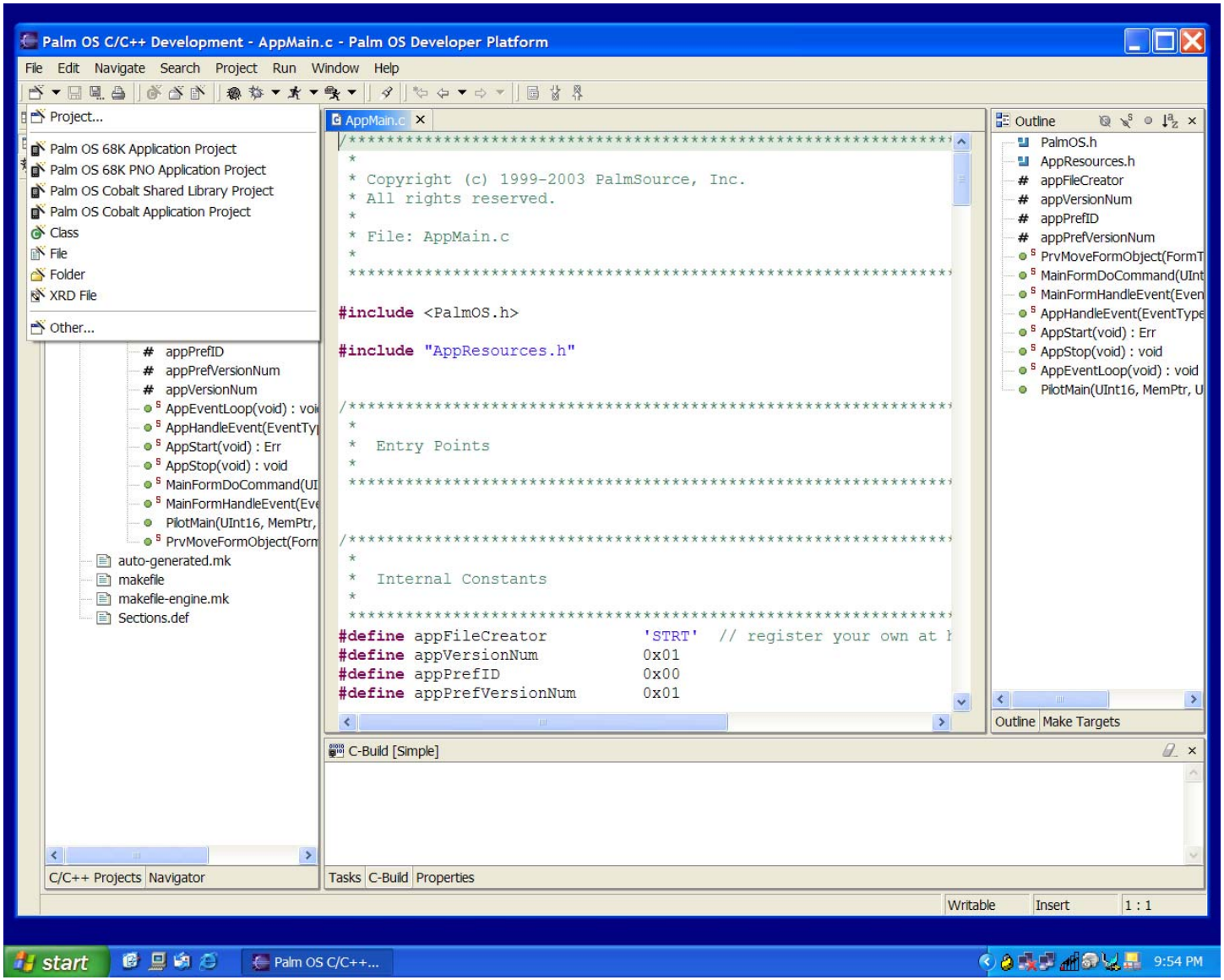
Integrated Development Environment

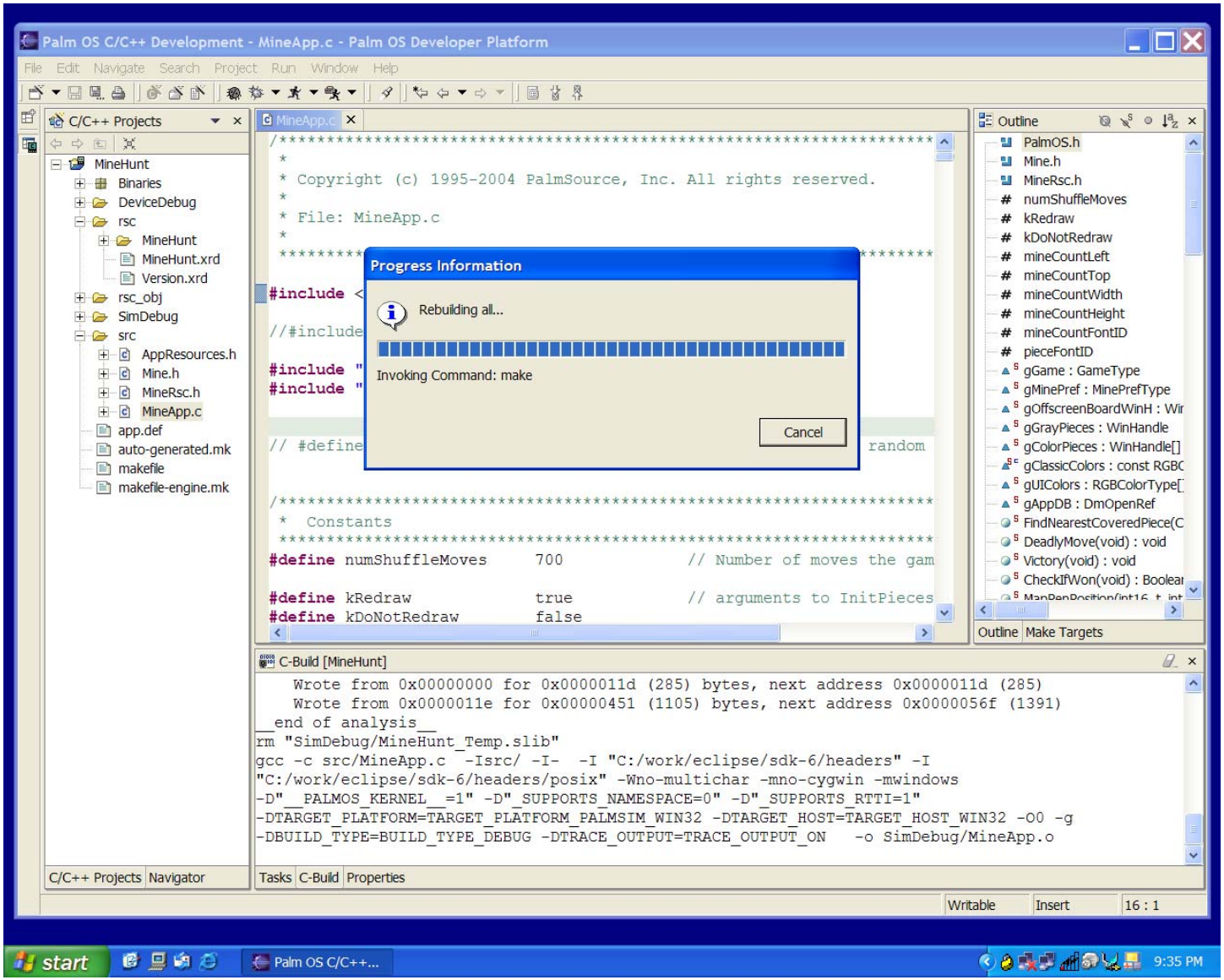
- **Palm OS® Developer Suite**
  - Based on Eclipse Version 2.1.2
  - C/C++ Development Tools Version 1.2
  - Plug-ins include:
    - Perspectives and Natures
    - Launch Configurations
    - Application Wizards
    - File Wizards
    - Help
    - Update Manager support



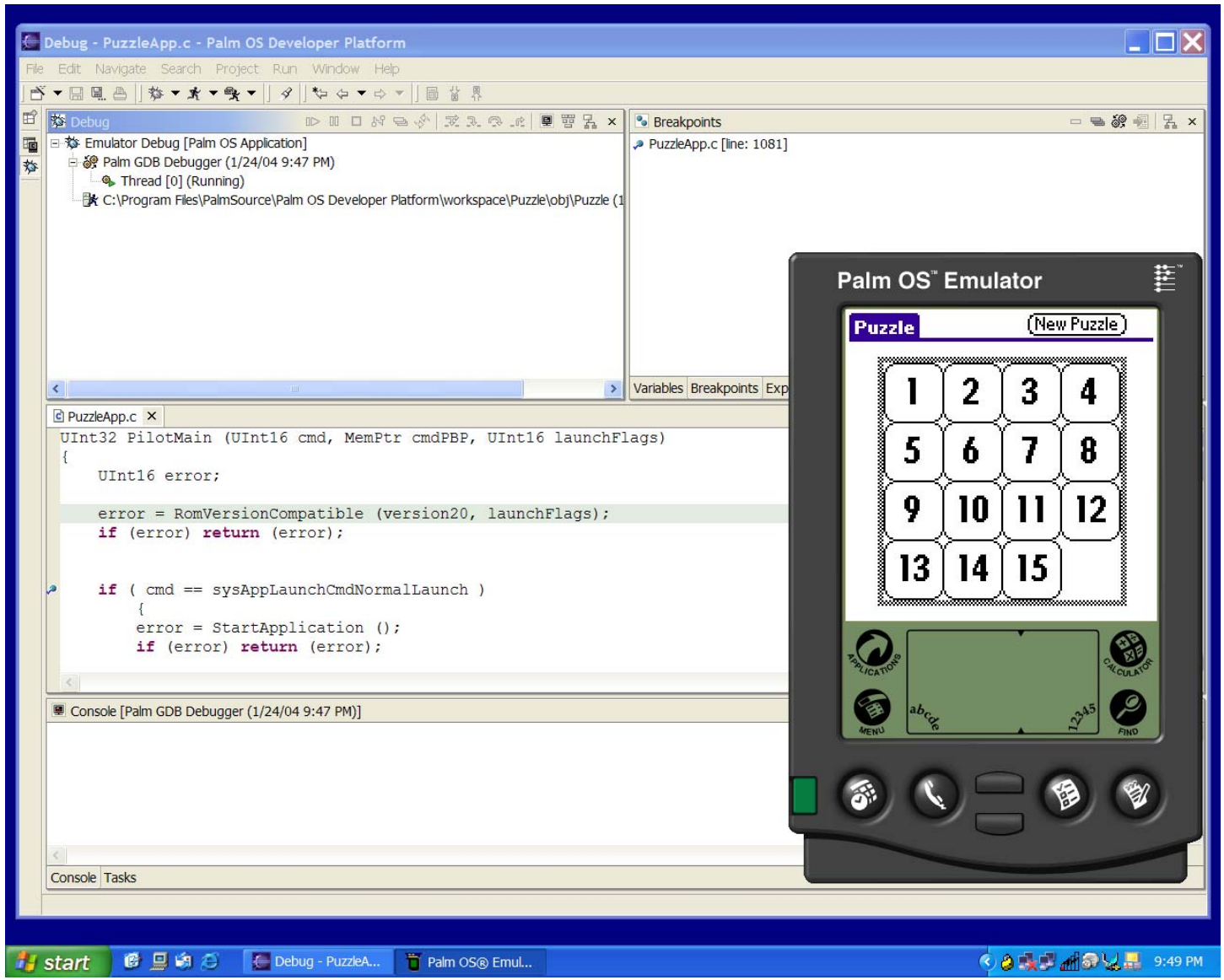
Integrated Development Environment

- **Palm OS® Developer Suite**
  - Cygwin and PRC-Tools
    - x86 Compiler and Debugger
    - 68K Compiler and Debugger
  - Palm OS Protein C++ Compiler
  - Palm OS Debugger
  - Palm OS Resource Editor
  - Palm OS Resource Tools
  - Palm OS Simulators
  - SDKs for OS5 and OS6
  - Standard Template Library
  - MathLib


Palm OS C/C++ Development - AppMain.c - Palm OS Developer Platform
File Edit Navigate Search Project Run Window Help
Project...
Palm OS 68K Application Project
Palm OS 68K PNO Application Project
Palm OS Cobalt Shared Library Project
Palm OS Cobalt Application Project
Class
File
Folder
XRD File
Other...
AppMain.c
* Copyright (c) 1999-2003 PalmSource, Inc.
* All rights reserved.
* File: AppMain.c
#include <PalmOS.h>
#include "AppResources.h"
* Entry Points
* Internal Constants
#define appFileCreator 'STRT' // register your own at h
#define appVersionNum 0x01
#define appPrefID 0x00
#define appPrefVersionNum 0x01
Outline
PalmOS.h
AppResources.h
appFileCreator
appVersionNum
appPrefID
appPrefVersionNum
AppStart(void) : Err
AppStop(void) : void
AppEventLoop(void) : void
PilotMain(UInt16, MemPtr, U
auto-generated.mk
makefile
makefile-engine.mk
Sections.def
C-Build [Simple]
C/C++ Projects Navigator
Outline Make Targets
Tasks C-Build Properties
Writable Insert 1 : 1
start
Palm OS C/C++...
9:54 PM


Palm OS C/C++ Development - MineApp.c - Palm OS Developer Platform
File Edit Navigate Search Project Run Window Help
C/C++ Projects
MineHunt
Binaries
DeviceDebug
rsc
MineHunt
MineHunt.xrd
Version.xrd
rsc_obj
SimDebug
src
AppResources.h
Mine.h
MineRsc.h
MineApp.c
app.def
auto-generated.mk
makefile
makefile-engine.mk
MineApp.c
Progress Information
Rebuilding all...
Invoking Command: make
Cancel
Outline
PalmOS.h
Mine.h
MineRsc.h
numShuffleMoves
kRedraw
kDoNotRedraw
mineCountLeft
mineCountTop
mineCountWidth
mineCountHeight
mineCountFontID
pieceFontID
gGame : GameType
gMinePref : MinePrefType
gGrayPieces : WinHandle
gColorPieces : WinHandle[]
gAppDB : DmOpenRef
DeadlyMove(void) : void
Victory(void) : void
Outline Make Targets
C-Build [MineHunt]
Wrote from 0x00000000 for 0x0000011d (285) bytes, next address 0x0000011d (285)
Wrote from 0x0000011e for 0x00000451 (1105) bytes, next address 0x0000056f (1391)
__end of analysis__
rm "SimDebug/MineHunt_Temp.slib"
gcc -c src/MineApp.c -Isrc/ -I- -I "C:/work/eclipse/sdk-6/headers" -I
"C:/work/eclipse/sdk-6/headers/posix" -Wno-multichar -mno-cygwin -mwindows
-D"__PALMOS_KERNEL__=1" -D"_SUPPORTS_NAMESPACE=0" -D"_SUPPORTS_RTTI=1"
-DTARGET_PLATFORM=TARGET_PLATFORM_PALMSIM_WIN32 -DTARGET_HOST=TARGET_HOST_WIN32 -O0 -g
-DBUILD_TYPE=BUILD_TYPE_DEBUG -DTRACE_OUTPUT=TRACE_OUTPUT_ON -o SimDebug/MineApp.o
C/C++ Projects Navigator
Tasks C-Build Properties
Writable Insert 16 : 1
start
Palm OS C/C++...
9:35 PM


THE PLATFORM FOR MOBILITY


Debug - PuzzleApp.c - Palm OS Developer Platform
File Edit Navigate Search Project Run Window Help
Debug
Emulator Debug [Palm OS Application]
Palm GDB Debugger (1/24/04 9:47 PM)
Thread [0] (Running)
C:\Program Files\PalmSource\Palm OS Developer Platform\workspace\Puzzle\obj\Puzzle (1
Breakpoints
PuzzleApp.c [line: 1081]
Variables Breakpoints Exp
PuzzleApp.c
UInt32 PilotMain (UInt16 cmd, MemPtr cmdPBP, UInt16 launchFlags)
{
UInt16 error;
error = RomVersionCompatible (version20, launchFlags);
if (error) return (error);
if (cmd == sysAppLaunchCmdNormalLaunch)
{
error = StartApplication ();
if (error) return (error);
Console [Palm GDB Debugger (1/24/04 9:47 PM)]
Console Tasks
Palm OS® Emulator
Puzzle
New Puzzle
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15
APPLICATIONS
CALCULATOR
MENU
FIND
abcde
12345
start
Debug - PuzzleA...
Palm OS® Emul...
9:49 PM



Palm OS® Protein C++ Compiler

- **Supports C and C++**
- **Industry-standard front end**
- **ARM 4T Instruction Set**
- **Includes tools set**
  - Compiler
  - Linker
  - Assembler
  - Shared Library Generator



Palm OS® Protein C++ Compiler

- **Where did it come from?**
  - Front End Licensed from Edison Design Group
  - Back End Licensed from Apogee
  - Engineered by PalmSource, Inc. for Palm OS
- **Testing**
  - In-house
  - Test suites are run constantly
    - Plum-Hall C and C++ test suites
    - STLPort C++ test suites



Palm OS® Protein C++ Compiler

- **Member of the ARM EABI Consortium**
- **Palm OS Cobalt is EABI-Compliant**

- **PalmSource Compiler Suite is…**
  - IA-64 C ABI Compatible
  - ARM C++ EABI Support
  - Interoperability with other tools



Palm OS® Protein C++ Compiler

Preliminary Performance Numbers

Dhrystone Numbers, Per Second

ADS	269,058.29
PSCS	260,869.56
GCC	258,620.68
CW	77,021.82

Dhrystone Numbers, Per Second (Signed/Unsigned Int Div)

PSCS	150,100.00
ADS	143678.01
CW	92,764.28
GCC	did not finish



Palm OS® Debugger

- **Standalone, Windows-based Debugger**
- **Debugs applications and shared libraries**
- **Supports debugging of…**
  - … ARM
  - … Thumb
  - … 68K
- **Source, assembly and mixed level debugging**
- **Designed to support multiple symbolic files and formats, including…**
  - ELF/DWARF 1
  - ELF/DWARF 2
  - SYM 3.3



Palm OS® Debugger

Use for 68K Development

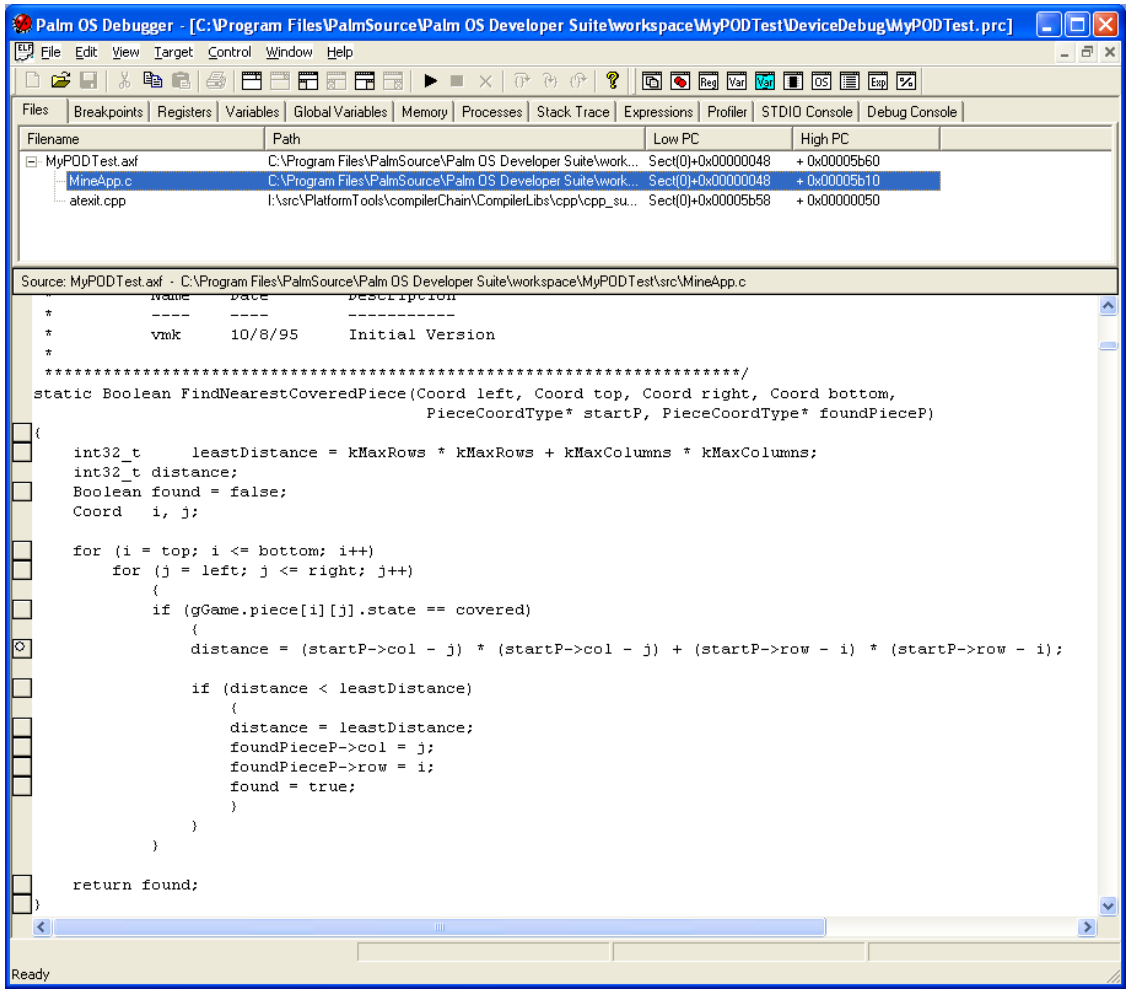
- **Palm OS Emulator**
- **Palm OS Simulators**
- **Palm OS Devices**

Use for ARM Development

- **Palm OS 5 PNO/68K Debugging**
- **Palm OS Cobalt native debugging**

(See Debugging Session, Thursday 11:30am)

Palm OS® Debugger





Palm OS® Resource Editor

- **Standalone, Windows-based Editor**
  - Property viewer
  - Visual editors
- **Visually edits XRD Files**
  - Alerts
  - Bitmaps
  - Forms
  - Menus
  - Strings



Palm OS® Resource Editor

- **Editors include**
  - Bitmap / Icon
  - Form
  - XML
- **Support multiple languages**
  - English, French, Italian, German and Spanish
  - Japanese
  - Traditional and Simplified Chinese
  - Korean
- **Partially integrated into our Eclipse release**
  - More integration forthcoming

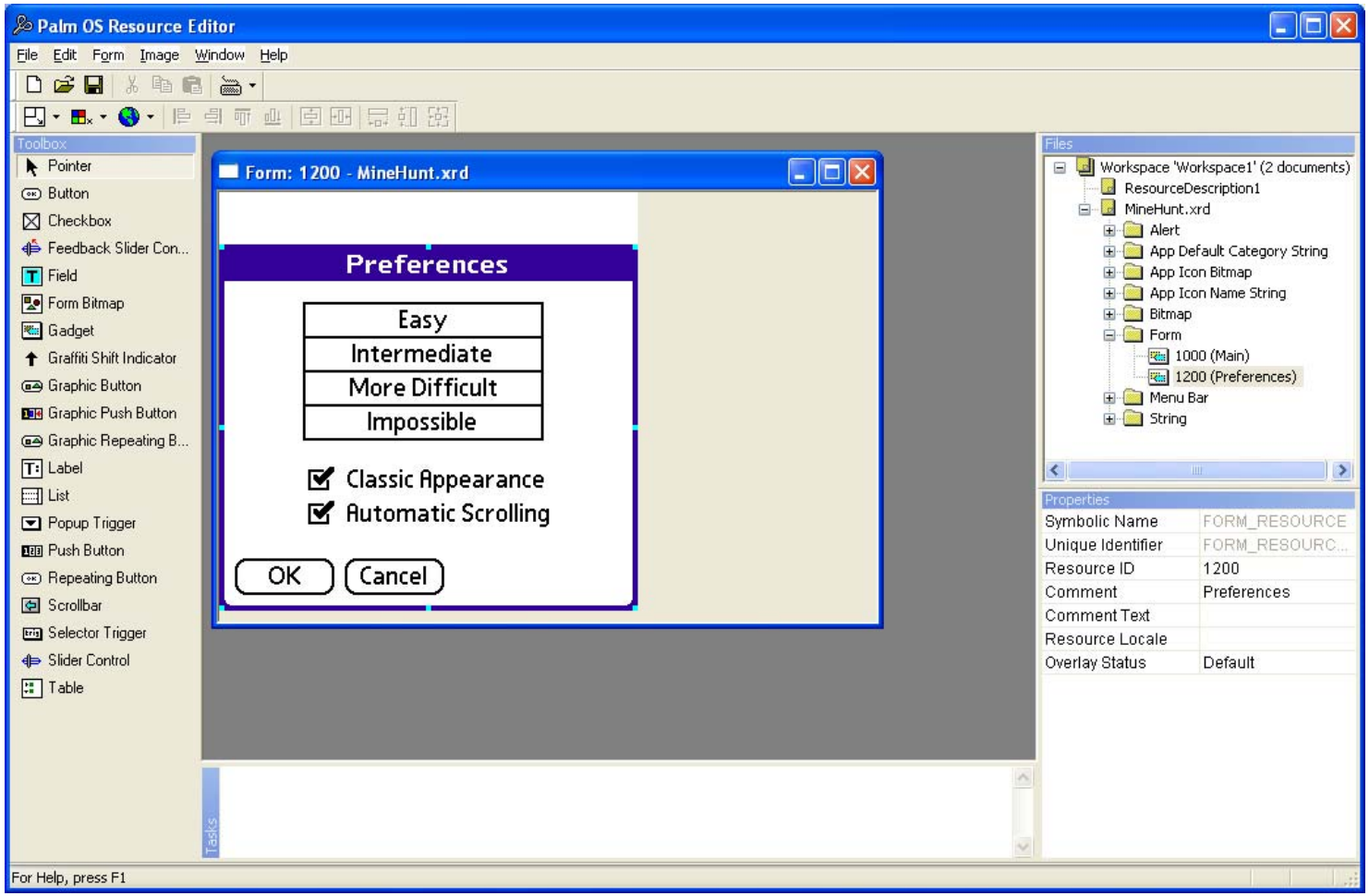


Palm OS® Resource Editor

What it is not

- **It is not cross-platform**
- **It is not extensible**
  - No extensions to language
  - No plug-in architecture
- **It is not a general XML editor**

Palm OS® Resource Editor





Palm OS® Simulators

- **Offspring of work done with the Palm OS**
  - It is closely coupled to OS releases
  - Most features existed in x86 first
- **The main tool used to develop Palm OS Garnet and Palm OS Cobalt**
  - Not everybody had development boards
- **Licensees have created variants**
  - Zodiac
  - T2
  - Treo 600
- **Integrated into our Eclipse release**

(See Simulator Session, Thursday 4:00pm)



Palm OS® 5 Simulator

- **PACE**
- **Viewers for**
  - Heaps
  - Databases
  - Events
- **Automated testing**
- **Adjustable color depths**
- **Expansion card support**
- **No skin support**



Palm OS® Cobalt Simulator

- **Run and test Palm OS Cobalt applications**
- **Run and test 68K applications (using PACE)**
- **New Features**
  - No notify reset
  - Selectable resolutions
  - Selectable screen sizes
  - Can specify HotSync name
  - Profiling
  - Threads View
  - Console

Palm OS® 6 Simulator





Palm OS® Cobalt Emulator

What can we expect?

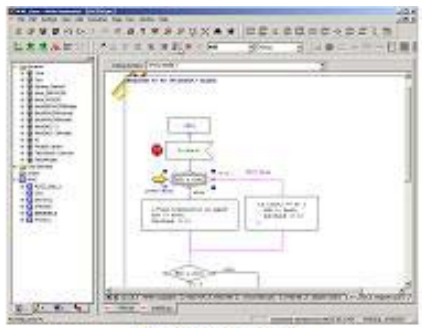
- **We are deciding on longer term strategies internally**
- **The Virtio emulator is available immediately**
  - "Generic" emulator available
  - Debugging support
  - HotSync support

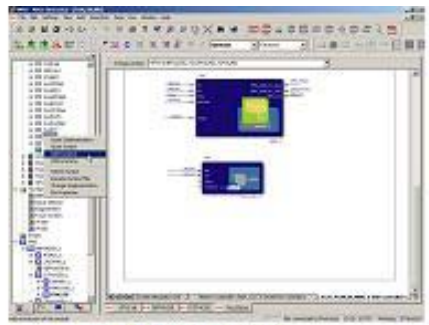
Virtio

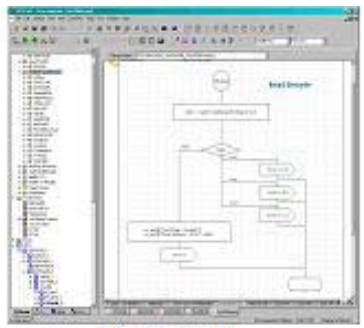
Technology Overview

- **Built from the ground up in collaboration with TI, Intel and Motorola**
- **Unique, fast simulation capabilities**
- **Unified hardware & software debugging**
- **End users receive custom "softboards" for fast and accurate device emulation**


Boot OS' in seconds


Breakpoint and
Single-Step in Hardware


Add components
From the library


Create your
Own models



PalmSource Emulation Roadmap

Proposed

Q1'04	Q3'04	Q1'05	Q3'05
Basic Emulation	Gremlins Support	Normal/Double size	IR Port Redirection
"Skin" Support	OS Monitoring	Saving of screen	3rd Level Dbg Options
HotSync Support	1st Level Dbg Options	Saving session	3rd Level Logging
Debugging Support	1st Level Logging	2nd Level Dbg Options	Profiling Support
Installing application	TCP/IP Redirection	2nd Level Logging	Hardware breakpoints
	HostFS Support	Tracing Support	



Conduit Development Kit 6.0

- **Still supports Visual Studio**
- **Still supports C++ and COM**
- **Access to the three types of databases**
  - Classic
  - Extended
  - Schema
- **Supports “folder” based registration of conduits**
- **Extended and Schema support in COM**
- **Desktop ships with COM APIs installed**

(See CDK/Desktop Session, Thursday 10:15am)



Java Overview

PalmSource licensed IBM's WebSphere Micro Environment (WME) Java 2 Micro Edition (J2ME) for Palm OS developers

- Supports industry standards
  - Connected Limited Device Configuration (CLDC) 1.0 and 1.1
  - Mobile Information Device Profile (MIDP) 1.0 and 2.0
- Both Palm OS Garnet and Palm OS Cobalt Java support are on the roadmap
- Developer runtime VM and Java development tools will be made available in the next few months
- Will work with licensees to develop VM runtime for different licensee devices
  - VM already available for Tungsten C devices from palmOne



Macintosh Development Tools

- **Macintosh has not been forgotten**
- **“Most” of our tools are Mac functional**
  - Compiler, linkers, assembler, resource compiler, etc.
  - Not: Resource Editor, Debugger or Simulator
- **Eclipse IDE**
  - Supports Macintosh development
  - C/C++ Development Tool
    - Does not *currently* support Macintosh
- **Interested in sampling the Mac wares?**

(Attend the Tools Roundtable tomorrow)



Future Plans

General

- **Developer bug-base for you to work with**
- **PNO Support in Palm OS Protein Compiler**
- **Macintosh releases of more tools**
- **Full integration of Palm OS Debugger**
- **Full integration of Palm OS Resource Editor**
- **Tighter integration with Emulators and Simulators**
- **More optimizations**
- **Managed Make facilities**



When

- **Palm OS® Developer Suite 0.1.0**
  - Technology preview available immediately
  - Expect updates through Update Manager shortly
    - 0.1.1 available at week's end
- **Palm OS® Developer Suite 1.0**
  - Available Q2/04
- **Free and Commercial Packages planned**
- **Palm OS® Developer Suite Updates**
  - Quarterly release plan
  - Opportunistic previews



What should you do?

Are you currently doing 68K development?

- Use your existing tools
- Learn the Palm OS Developer Suite

Are you currently doing PNO development?

- Use your existing tools
- Learn the Palm OS Developer Suite

Are you interested in doing Palm OS Protein development?

- Use Palm OS Developer Suite
- Stay tuned for updates
- Attend the labs, email concerns/bugs, harass my engineers!



Next Steps

- **Incremental releases of *Palm OS® Developer Suite***
- **"Thrillseeker" releases of development tools**
  - PNO support in compiler
  - Frequent releases of updates/fixes
  - Opportunistic releases of new functionality

Visit the Lab at 7:30pm tonight for help on using the tools and implementing your first application



Tools Team: Introductions

- **Eclipse Team**
  - Brad Jarvinen
  - Keith Rollin
  - Mahmood Sheikh
  - Ray Pierce
- **Compiler Team**
  - Kevin MacDonell
  - Jeff Westerinen
  - Kenneth Albanowski
- **Resource Team**
  - Eric Cloninger
- **CDK Team**
  - Ravi Duggaraju
  - Cole Goeppinger
- **Debugger Team**
  - Greg Clayton
  - Steve Lemke
  - Matt Fassiotto



Questions and Answers?

phil.shoemaker@palmsource.com

408-400-1599



palm OS® Cobalt

Contact: Doug MacMillan
Sr. Palm OS Platform Evangelist
Email: doug.macmillan@palmsource.com

Contest: Open Labs (Tues/Wed)
68K Compatibility & Protein Contests
Motorola sponsored Grand Prize!

MOTOROLA
digital dna